

Strengthening Systems of Support to Improve Early Childhood Development**September 2026****Working Paper**

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Despite evidence that the Supplemental Nutrition Assistance Program (SNAP) improves children's health, education, and long-term economic outcomes, take-up is incomplete among eligible populations. While prior research shows that ethnic-peer networks can facilitate program participation, less is known about whether networks formed around shared need or interest shape access. This paper examines whether the availability and concentration of resource institutions (churches, preschools, elementary schools, and food banks) is associated with SNAP take-up among young children in North Carolina. Leveraging shifts in institutional resources over time, I use marginal structural modelling with inverse probability of treatment weights to estimate the relationship between SNAP take-up and the concentration of resources and distinct institutions. Results show that a greater number of preschools, elementary schools, and churches per child is associated with higher SNAP take-up. These higher per-capita resource counties tend to be smaller, with fewer aggregate resources overall but higher concentrations of poverty, suggesting that smaller counties with more concentrated, accessible resources may be better positioned to connect eligible families to public assistance. These findings extend the safety net access literature by identifying community resource institutions as a mechanism to connect families to systems of support and point to targeted partnerships between local agencies and organizations as a promising strategy for improving program access and child welfare in under-resourced areas.

Keywords: safety net access, community resources, early childhood, brokers

Poverty can have long lasting effects on one's trajectory, and children who face poverty for extended periods of time, and during their early years, are more likely to be impacted (Bradley et al., 2001; Brooks-Gunn & Duncan, 1997; Chaudry & Wimer, 2016; Duncan et al., 2011; Evans, 2004). With increasing disparities in income equality across the United States (US), there are widening gaps in attainment and economic mobility among children from lower and higher income families (Duncan & Murnane, 2016). This is not an individual level problem. Changing children's outcomes first requires society to reduce the barriers to opportunity and strengthen the channels for children to receive supports aimed to mitigate the harmful effects of poverty.

Early intervention through educational and social safety net programs offer promising policy solutions to mitigate the adverse effects of poverty (Dreyer, 2019; Heckman & Masterov, 2007). The federal Supplemental Nutrition Assistance Program (SNAP) has been found to improve children's health, nutrition, economic, and education outcomes (Frongillo et al., 2006; Gassman-Pines & Hill, 2013; Hoynes et al., 2016). Research further demonstrates that children with access to SNAP during early childhood show improvements in health and greater economic stability into adulthood (Hoynes et al., 2016). Collectively, this evidence suggests that providing children with essential supports during critical developmental periods can produce lasting benefits for educational attainment and long-term economic mobility.

Beyond household food, participating in SNAP offers cascading benefits through schools. When families enroll in SNAP, students can access free meals, which also spurs additional school funds through Title 1 and state funding formulae targeting vulnerable children. School meals positively impact academic achievement (Schwartz & Rothbart, 2020) and behavior in school (Domina et al., 2024). The positive benefits of increased school funding for children whose households face hardship have also been well documented (Jackson & Mackevicius, 2024). SNAP benefits participants through both individual food support and additional supports in the school setting.

SNAP's promise hinges on participation. The take-up rate among eligible persons is just over 60% (Macartney & Ghertner, 2021). Enrollment is key to observing these compounding effects and improved long-term outcomes. There are both individual and programmatic barriers that affect participation. Key factors in access are knowing eligibility status, knowing how and where to apply and reducing the stigma (Herd & Moynihan, 2019; Homonoff & Somerville, 2021). Individuals can overcome these barriers; yet networks can also play a key role in supporting families.

Having a network who can provide details or encourage a family to apply may be useful in improving take-up for federal programs like SNAP. These networks can be established through shared interests and goals or proximity. Current research shows that living near same ethnicity peers is associated with increased program take-up (Bertrand et al., 2000). There is less known about how networks related to shared interests or need are related to take-up.

This paper examines whether the availability and concentration of resource institutions are connected to SNAP take-up. Using North Carolina (NC) data, I examine whether certain institutions are a key player in improving access to federal programming. To examine these relationships, I use linked administrative data from North Carolina's Longitudinal Data Service, Department of Health and Human Services (DHHS), and Department of Public Instruction (DPI). The DHHS data provide a census of SNAP participants from birth through age five, while the DPI data capture broader population information for children in kindergarten through third grade. These data are combined with Census Bureau's American Community Survey (ACS) data, which estimate the population of eligible children for SNAP. Together the participants compared to the eligible population represents the take-up rate. I track the location and operating status of community resources from 2016 to 2019 to assess whether the concentration of institutional resources within communities relates to SNAP take-up and access to public assistance more broadly.

This evidence extends the safety net access literature to examine community-level factors related to gaps in participation. Using county-level measures, I evaluate whether increased take-up is connected to the institutions in the county that children are likely to use. These institutions include churches, preschools, elementary schools, and food banks. These findings provide insights on the mechanisms that connect families to broader systems of support that promote child welfare. With information on the institutions within a community, interested parties can create more targeted approaches to strengthen partnerships between agencies and community organizations within to better reach individuals.

Results indicate that having more preschools, elementary schools, and churches per child is related to increased SNAP take-up. In NC these higher per capita resource areas are the smaller counties with less aggregate resources overall and higher concentrations of poverty. In the NC context, this evidence suggests that smaller counties with smaller quantity, but larger per-capita resources may be better situated to meet the needs of their members and improve access to available programming to promote child welfare. This evidence highlights the importance of community resources, specifically in areas where there is less access to opportunity and resources. Ensuring that families are connected to these resources and that agencies partner with them may be a pathway to ensure families can access the resources needed to mitigate the effects of poverty and promote child welfare.

Background

SNAP Access

SNAP is a federally funded, county administered, entitlement program for families experiencing economic hardship. Families with children are eligible to receive benefits if their income is below 130% of the federal poverty level (FPL)¹. NC expanded access through broad-based

¹ Families without children have additional work requirements for eligibility.

categorical eligibility, so families are eligible up to 200% FPL. In NC, families can enroll through an online application, in person at the Department of Social Services office, or by mail. Each county has its own office that oversees applications and distribution of benefits. Families can only apply through these means, so each county can be considered as having its own process or shared experience.

The Role of Community Resources

Institutions cannot enroll families in SNAP; yet they can encourage (or deter) enrollment. Families are engaged with most of these institutions out of necessity. They need childcare and attend preschools, have children that are attending elementary school, congregate at churches based on religious beliefs, or may need food assistance from food banks. Connection to community institutions allows families to learn about available resources and share institutional knowledge with other families about the processes to gain access (Jencks & Mayer, 1990; Small, 2006). These routine organizations are the primary channel to form social ties outside of one's familial network, especially for families from lower income communities (Small & Gose, 2020). Families meet based on a shared need and become tied to the same network.

These community institutions then serve as information channels for families. They are considered “brokers” because they can connect families to additional resources (organizational brokerage) or they connect families to other families that share resources (individual brokerage) (Small & Gose, 2020; Stovel & Shaw, 2012). Families can share their experience with the application process and the eligibility information with other families within their network. These institutions congregate larger and more diverse groups than one's immediate network, which increases the information available to families.

As families build trust with community institutions and the members, they may be more inclined to share their hardship with others. Because of the stigma related to needing support, families may not share their need with their close network or kinship. It may be easier to converse with people who share weak ties through these organizations (Small, 2006). Both the institution and members can share information about supports, and these institutions can use their social capital to connect families to broader systems of support (Warren, 2005). As there may be more trust within these communities, connection to networks may be an important factor to improve outreach and access to larger systems of support. This paper tests this relationship to determine whether these community institutions do in fact increase access to the social safety net.

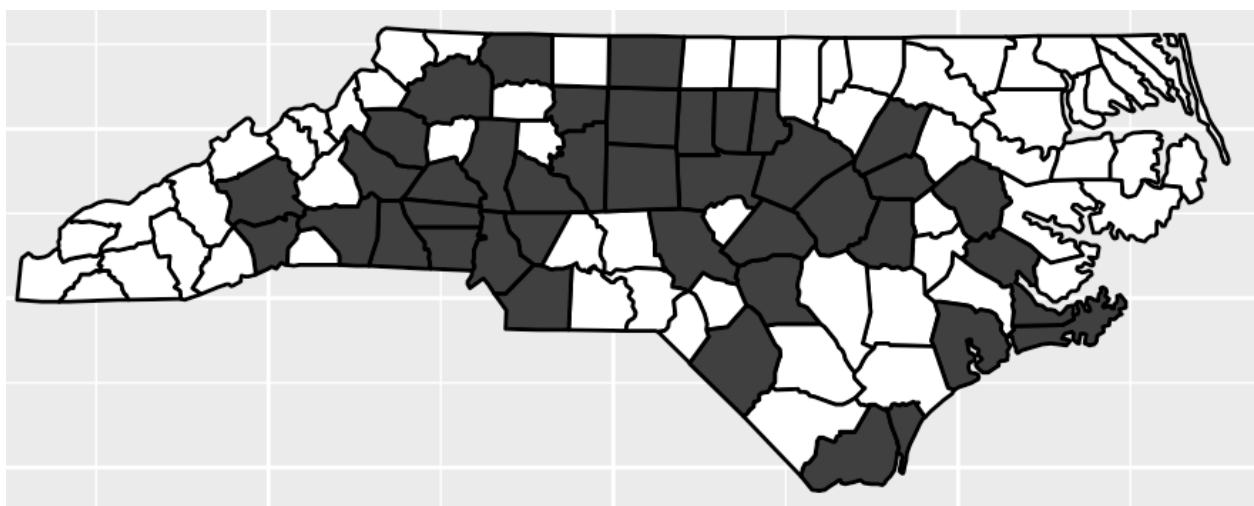
Data

To examine take-up, information is needed on both participants and eligible non-participants. State administrative data from DHHS documents all participants in SNAP; however, agencies are only able to see who participates. To establish an eligible population, I supplement these data with ACS data. The 1-year ACS estimates the number of children within a county by income to poverty thresholds. To align with NC's eligibility of 200% FPL, I use the number of children at

or below 199% FPL. The 1- year ACS surveys areas with a population of 65,000 or greater. These estimates provide coverage for 40 NC counties (*See Figure 1*), representing 80% of the state population.

I construct a panel spanning from 2016 to 2019. Each county has a take-up rate, which is defined as the number of children under six who participate compared to the eligible population. A rich set of county covariates is used from ACS and US Department of Agriculture to account for the county characteristics related to take-up and community resources.

Figure 1.
Counties with 1-Year ACS Estimates



Note: Counties in the sample are shaded gray

In addition to county level analysis, the NC administrative data allow for individual-level analyses. For this group of children who participate in SNAP at any point, I can link them across additional state program data. Children may not participate in SNAP for all five years of early childhood. However, for some children, they still show up in the data from participating in other early childhood programs. These programs have income eligibilities aligned with SNAP, so eligibility can be established. This provides a panel of 116,500 children who have data spanning 2016 – 2019.

Empirical Approach

Measuring Community Resource Levels

Community institutions are the social institutions within an area that provide essential resources to its inhabitants and buffer families from the harmful effects of poverty (Wilson, 1997). One mechanism may be through the connection to broader systems of support by serving as

information brokers for these families. And because families experiencing economic hardship are more likely to rely on community resources, these institutions are important social ties for this eligible population (Small & Gose, 2020).

For these resource institutions to factor into SNAP take-up, they must be resources that families are likely to use during early childhood. As discussed above, families must engage with these resources on a frequent basis to share and receive information. Based on this assessment, I include preschools, elementary schools, food banks, and churches. The location and operation status of these organizations are available through Data Axle and NCES. Data Axle is a historical database of businesses in the US used to identify many of these institutions. These data are available through the *Wharton Research Data Services*. National Center for Education Statistics (NCES) Common Core Data (CCD) identify all K-12 public schools.

Consistent engagements with these organizations require them to have capacity to meet the needs of the members. To operationalize this process, I account for the resources per child for each organization type. Each unique business is counted once within a county and a year, and the sum is then divided by the number of children under 6 within the county. This measure is defined as the resources per child. Analyses are done separately for each resource and with the total of all resources to determine if there is a relationship between the concentration of resources, individual institutions, and SNAP take-up.

Families are only likely to engage with one of each of these resources, and there is a greater likelihood for participation closer to their neighborhoods. In addition to the county-level measure, I create a city-level measure that is then aggregated to the county. I identify the number of resources in a city within the county and average across the cities within that county to account for the variation in access due to location within a county.

These resource measures account for the per capita resource, but it may be a closer representation of capability over capacity of an organization. To target capacity, I use measures specific to the institutions. For preschools, I have access to the number of employees at each site, which I divide by the maximum class size. In NC, regulations cap younger preschool classrooms at 18, and PreK classrooms at 20. Using the average, capacity is defined as the number of employees per max classroom size. For elementary schools, capacity is the ratio of the number of full-time equivalent teachers to the school enrollment. For churches, I use the same concentration measure because attendance is about religious beliefs and finding the best location, so the concentration captures this capacity. Food bank capacity is the yearly revenue compared to the number of people below the FPL.

The resource level is calculated for each county. At the county level, it is easier to interpret differences between lower and higher resourced counties rather than a continuous measure. I create a categorical measure and group counties into quintiles for each year to assess the difference across lower and higher resourced counties. This grouping is based on the yearly distribution of resources across the counties. This is useful in the modeling approach and

interpretation, which is exploring the relationship between living in a lower or higher resourced community over time.

Identification Challenges

Families are not randomly assigned into counties, and these decisions can impact access to opportunity (Chetty et al., 2016). Without accounting for this selection, estimation of the effect of community resource levels on take-up would yield biased results that may be related to neighborhood or family characteristics. To address this endogeneity, policy changes related to SNAP or community institutions could be leveraged. During this window, there were no policy shifts. SNAP policy for this population has remained stable since the expansion of broad-based categorical eligibility. In the absence of these shifts, I rely on panel data approaches to leverage the longitudinal shifts within the data.

Modeling Approach

Previous studies have examined the effects of neighborhoods on children's education. Sampson et al. (2008) find that concentrated neighborhood disadvantage has a negative effect on literacy and vocabulary skills. Wodtke et al. (2011) find that the duration of exposure to neighborhood disadvantage has an effect on the likelihood of graduating from high school. I follow their methodological approach and use marginal structural modeling (MSM) with inverse probability of treatment weighting (IPTW; Hong & Raudenbush, 2008; Robins et al., 2000) to account for selection into neighborhoods and the serial correlation with confounders.

MSM leverages dynamic treatment regimes with time-dependent covariates. Dynamic treatment allows for estimation of the shifts in levels over time. Weighting by confounders that are also shifting over time removes the association between the county and the dependent variable to examine the relationship between take-up and resource levels. Neighborhoods are defined by the interaction between multiple characteristics. This modeling approach accounts for the confounders simultaneously instead of controlling for each separately. MSM also accounts for the serial correlation between the confounders and measures the relationship based on the level of exposure to a specific treatment.

2016 serves as the baseline with 2017 being the first year of treatment. During this window, counties are gaining and losing resources, which shifts their resource concentration. A county's resource level per child is changing across time (*See Appendix*). In addition, these changes also lead to shifts in the county's place in the distribution each year, across the five levels (*See Figure 2*). Families are also moving and changing locations during this time frame. One-fourth of children move at least once during the panel.

Estimation of the shifts in resource level would still lead to biased results, because there are certain county-level characteristics that are related to resource concentration and take-up, and these characteristics could influence later outcomes. For example, persistent poverty is associated with a decrease in public services within a community (Blank, 2005). This suggests

that how these characteristics change or persist over time is also an important factor alongside the present rate. In addition, these neighborhood characteristics are intertwined (Sampson et al., 2008), suggesting it is not one sole factor, but multiple factors together that characterize neighborhoods. Instead of controlling for these measures, I use IPTW to account for the serial correlation and to consider all confounding factors collectively rather than holding each constant.

This modeling approach creates a pseudo-population that is weighted by time-varying confounders and past resource levels and take-up to remove the connection between county characteristics and resources. Observations are weighted by the probability of the observed county resource level conditional on past resource level, take-up, and county characteristics (Robins et al., 2000; Wodtke et al., 2011). Estimation accounts for the past treatment and determines the probability of the observed outcome based on the observed trajectory to weight observations. Weighting adjusts the observed data so observations with all levels of exposure across time are equally represented among all levels of the confounders. In the final model, observations with a high probability are down-weighted and those with low probabilities are up-weighted. Because there are many covariates related to resource levels and SNAP take-up, this weighting approach also reduces the dimensionality of the variable to avoid curse of dimensionality challenges.

While this empirical approach reduces bias in estimation, these estimates are still considered correlational. MSM requires that all confounding factors are accounted for, and although I consider many county characteristics that may be confounders, there is a possibility a Z remains unaccounted for. Another consideration may be fixed effect modelling; however, MSM may be better suited when examining community factors. Fixed effects modelling assumes the unobserved group effect, c , is constant over time. In the case of neighborhoods, this assumption is unlikely to hold. MSM accounts for the changes in the variables over time and across all levels, reducing the bias in estimation.

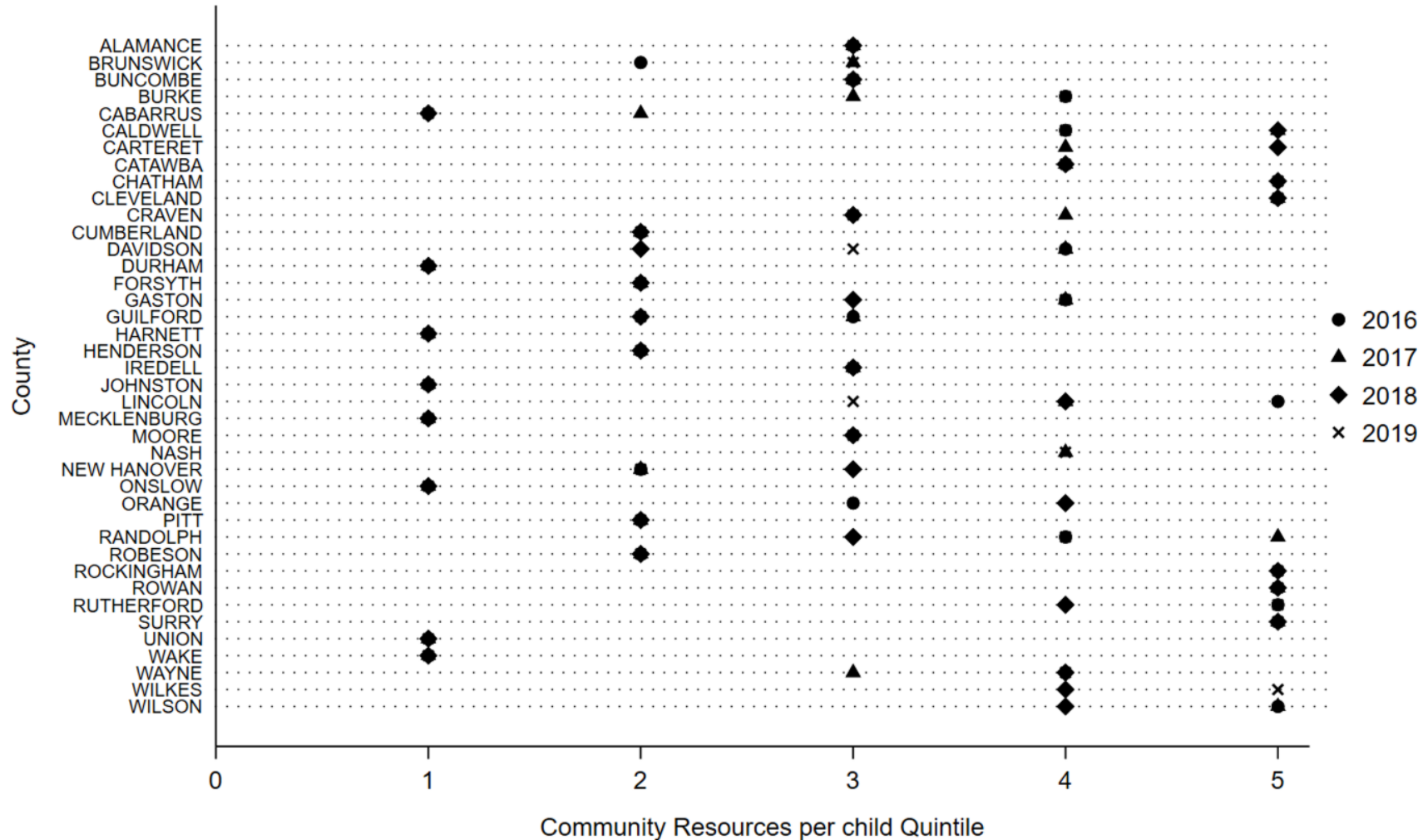
Model

Each county, at time t , has a set of potential outcomes based on the prior levels of resource concentration up to that point, and the characteristics of the county, both fixed and time-varying (Holland, 1986; Robins et al., 2000). In these data, counties are classified by their resource level, which is a categorical variable representing the county quintile. I estimate the relationship of a certain resource concentration,

$$E(Y_{\bar{a}} - Y_{\bar{a}'})$$

which represents the difference in exposure sequences over time. $A_k \in \{1, 2, \dots, 5\}$ represents county exposure status at the k th year after baseline, and $\bar{a}$ represents a certain exposure trajectory. Y_{ct} is the SNAP take-up rate defined as the number of SNAP participants compared to the number of eligible children under 6 for a county, c , at time, t . At the individual level, it is the likelihood of participating in SNAP for an individual, i , in county, c , at time, t .

Figure 2.
Community Resources per Child as a Categorical Measure, by Year



Note: Resource Level is defined as the sum of all resource organizations compared to the number of children under 6. Levels were created from the yearly distribution. *Source:* Business locations are from Data Axle and NCES Common Core

The first step is to create the pseudo-population using IPTW. I obtain parametric estimates of exposure probability by estimating the $p(A_t | H_t)$, which is the probability of the observed resource concentration level, conditional on all history, H_t . H_t includes past concentration levels, past take-up, and county covariates. In this pseudo-population, the probability of treatment no longer depends on the confounders, L . To improve estimation, I use stabilized weights.

$$sw_c = \prod_{t=1}^T \frac{P(A_t = a_{ct} | \bar{A}_{t-1} = \bar{a}_{c(t-1)}, L_0 = l_0)}{P(A_t = a_{ct} | \bar{A}_{t-1} = \bar{a}_{c(t-1)}, \bar{Y}_{t-1} = \bar{y}_{c(t-1)}, \bar{L}_t = \bar{l}_{ct})}$$

Stabilized weights estimate the ratio of the probability of treatment status conditional on baseline covariates against the probability of treatment conditional on time-varying covariates. Estimation using stabilized weights have a smaller variance and an approximately normal sampling distribution (Hernán et al., 2002; Robins et al., 2000). This approach accounts for county and past exposure trends to reduce the bias in estimation of the relationship between community resource levels and SNAP take up.

Results

County Characteristics

The analysis sample includes 40 counties across NC. 60 of NC counties have a population below 65,000, so there are no yearly ACS estimates to determine the SNAP-eligible population of children under 6. In addition, these counties' resources levels remained largely stable over the study period (*See Appendix*). The identification strategy relies on within-county temporal shifts in resources, so these counties do not meet the criteria for inclusion. Future work should explore these smaller and rural populations to understand how resources relate to take-up with more localized data for these populations.

In the analytic sample, the average poverty rate is 14.65%; the childhood poverty rate is much higher with an average of 25%, as shown in *Table 1*. In these 40 counties, nearly 50% of children, on average, are eligible for SNAP, based on the 200% FPL income threshold. The analytic sample spans counties ranging in population from 65,000 to 1.1 million, indicating substantial variation in region size. As such, these findings may hold relevance beyond large urban areas alone.

The average take up is 75% across these counties; in certain counties it is as low as 37%, suggesting there may be a gap in participation for certain areas. The take-up rate ranges both by county and year, with some counties largely stable and others experiencing large variation across time (*See Figure 3*). While overall participation has declined over time, this trend varies by county, underscoring the need for a deeper understanding of the place-based characteristics associated with take-up.

Community Resource Institutions

The average community resource level is .028 per child. Churches are the most prevalent resource. The resource per child measure accounts for the total number of resources and the number of children that would use these resources. In NC, those with the highest resource per-capita, are the counties with the lowest number of individual institutions. As shown in Table 2, counties in the first level have the largest populations and the highest quantity of each resource. These counties also have the fewest eligible children and the lowest SNAP take-up rate, compared to other resource levels. The overall resource measure and the individual resource-specific variables do not fully overlap; counties may rank in the lowest category on one measure, while ranking higher on another. This pattern indicates that the measures capture distinct dimensions of resource availability, supporting their use as separate indicators rather than duplicate measures for a single construct. Accordingly, both the composite and resource-specific measures are used in the analysis to capture overall resource context as well as variation across individual resource institutions.

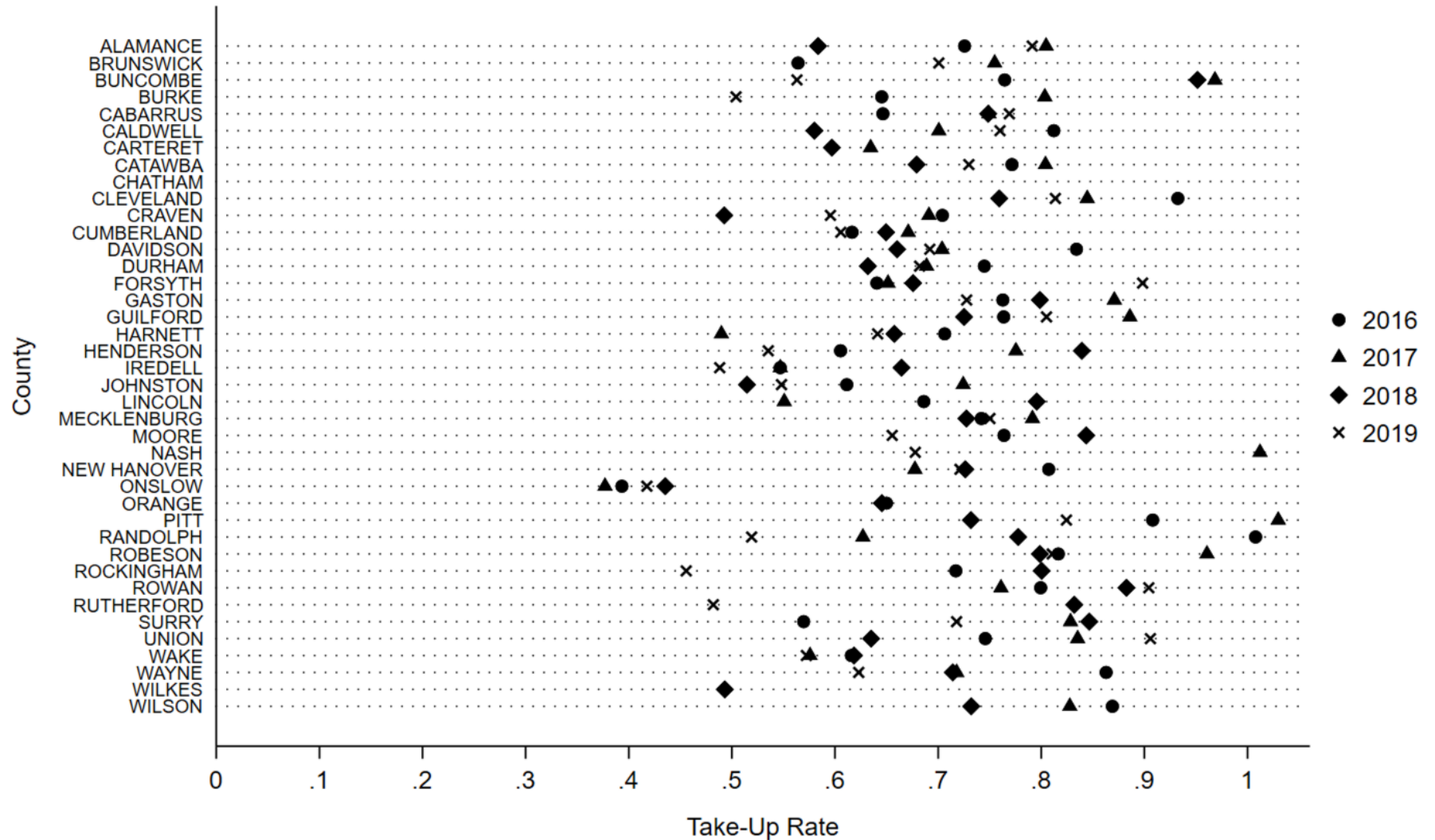
Table 1.
Descriptive Statistics of Sample

Variable	Mean	Standard Deviation
Take-up	0.71	0.13
Resources per Child	0.03	0.01
Elementary Schools	27.51	23.34
Preschools	63.66	67.67
Churches	233.60	153.46
Food Banks	1.61	0.88
Community Centers	7.93	9.81
Preschools, City	10.24	9.23
Community Centers, City	1.71	0.99
Food Banks, City	1.43	0.66
Churches, City	21.37	15.43
Elementary Schools, City	3.85	3.12
Children under 6	16,188.33	18,193.40
Population	222,053.33	231,040.67
Unemployment Rate	4.47	0.91
Black	0.20	0.11
Hispanic	0.09	0.03
Education, no diploma	0.13	0.04
Children below 199 FPL	0.51	0.13
Children below FPL	0.25	0.08

Note: 40 counties are included in the analytic sample from 2016 - 2019.

Source: Participation data from NC DHHS. County Characteristics from ACS and USDA.
 Business Counts from Data Axle and NCES Common Core Data

Figure 3.
SNAP Take-Up for Children Under 6, by County and Year



Note: The take-up rate is the SNAP participation children under 6 compared to the eligible population

Source: Participation numbers are from the NC DHHS. Eligible children are from the ACS.

Table 2.
Group Differences by Community Resource Quintile

	Community Resource per Child Quintile				
	1	2	3	4	5
Take-up	0.64	0.74	0.72	0.73	0.74
Resources per child	0.02	0.03	0.03	0.03	0.04
Elementary Schools	47.23	33.00	24.13	18.93	15.33
Elementary Schools, City	6.81	4.33	3.27	3.00	2.25
Preschools	108.45	70.69	49.04	35.71	21.50
Preschools, City	15.57	13.35	7.56	6.09	4.80
Churches	323.71	274.11	218.53	180.71	151.19
Church, City	32.59	23.39	18.43	17.44	12.98
Children under 6	33,932.90	17,391.61	11,973.97	8,205.14	5,839.30
Population	439,374.55	243,307.68	184,737.53	125,619.43	91,962.37
Unemployment Rate	4.23	4.78	4.19	4.49	4.70
Black	0.24	0.25	0.16	0.17	0.15
Hispanic	0.12	0.09	0.08	0.08	0.08
Education, no diploma	0.11	0.12	0.11	0.16	0.17
Children below 199 FPL	41.12	53.28	46.40	54.54	60.74
Children below FPL	18.72	28.63	21.85	27.41	29.45
Preschool Quintile	2.03	3.62	3.12	3.00	3.32
Church Quintile	1.07	2.14	2.97	4.00	4.82
Elementary Quintile	1.26	2.54	2.87	3.75	4.56

Note: Data for 40 NC counties from 2016 - 2019. All group differences significant at the $p < 0.05$ level.

Take-up and Community Resources

When using a continuous measure, having more preschools and elementary schools is connected to increased take-up. The average resource level is 0.004 for preschools and 0.002 for schools. Increasing the concentration by .001 is associated with a 6% increase in take-up for preschools, and a 7.5% increase for elementary schools.

Another approach is to group counties by their resource amounts each year and create levels. This eases interpretability to consider going from a lower-resourced to a higher-resourced county. Quintiles are created based on the yearly distribution for each measure. The model estimates the relationship between moving from the lowest-resourced area to each additional level.

Moving from a county with the lowest level of resources per child to the next level is associated with 11.5% increase in take-up. Elementary schools and churches were also connected to an increase in take-up across multiple levels. There was no relationship between preschools at the county level, but having more preschools per child within the city is associated with increased take-up. Moving from the lowest-resourced area to the next level is associated with a 10% increase in take-up. The preschool findings highlight the importance of access within more granular regions, signaling the choice options within closer proximity to residence.

None of the individual institution capacity measures were significant; yet moving from the lowest-capacity area to the second level area is related to an 8.5% increase in take-up. Analysis could not be conducted for food banks due to sparsity within the regions.

Overall, these findings suggest that having more resources per child is connected to improved SNAP take-up. In the context of NC, the counties with higher resources per child are areas with less institutions overall and with smaller populations. This suggests that these smaller counties are better able to reach their members and connect them to services. This may be because resource institutions in areas of disadvantage and less opportunity are critical bridges for their members, who may be more difficult to reach by outsiders (Small & Gose, 2020). In addition, with fewer overall options, these communities may be better positioned to reach and engage residents effectively.

This evidence underscores the potential power of community-level connection and cohesion. Even amid economic disadvantage, closer proximity to available resources, and stronger ties within the community, may meaningfully enhance program reach and engagement. These findings suggest that the structure and accessibility of resources within a community, rather than volume alone, may be a critical factor in facilitating take-up. Further research is needed to examine these networks of resources at a more localized analysis level to explore the mechanisms that drive this relationship.

Table

Connection Between Take-up and Community Resource

Association Between SNAP Take-up and Resources, by Resource Organization			
Resources per child	Resources per child, City	Resources per Child, Capacity	
1 to 2	0.115*	0.059	0.085*
	0.053	0.060	0.038
1 to 3	0.095	0.075	0.045
	0.054	0.062	0.044
1 to 4	0.074	0.040	0.012
	0.047	0.056	0.054
1 to 5	0.112	0.009	0.033
	0.057	0.059	0.041
Preschools	Preschools, City	Preschools, Capacity	
1 to 2	0.095	0.101*	0.030
	0.069	0.038	0.052
1 to 3	0.043	0.105	0.025
	0.069	0.053	0.070
1 to 4	0.087	0.010	0.044
	0.067	0.051	0.059
1 to 5	0.043	0.039	0.020
	0.063	0.044	0.056
Churches	Churches, City	Same Model as Churches	
1 to 2	0.143**	0.102*	
	0.051	0.050	
1 to 3	0.067	0.071	
	0.053	0.062	
1 to 4	0.053	0.018	
	0.045	0.052	
1 to 5	0.116*	0.030	

	0.050	0.050	
Elementary School		Elementary School, City	Elementary, Capacity
1 to 2	0.073	0.069	0.012
	0.043	0.057	0.061
1 to 3	0.166**	0.038	0.031
	0.051	0.065	0.059
1 to 4	0.102	0.061	0.025
	0.052	0.057	0.066
1 to 5	0.099*	-0.002	0.025
	0.049	0.056	0.064

*Note: All models include controls for county covariates: unemployment rate, Black %, educational attainment no high school diploma, population school-aged %, and SNAP county budget. Other characteristics were excluded when correlation was higher than 0.6. Robust standard errors. Institution models were run separately. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$*

Data Limitations

Larger counties are classified as lower resourced per child, but some of this may be due to the analysis level. These larger counties have more variation in resources across regions within the county. The study measures are at the county level, so I am unable to detect these differences. A more fine-grained analysis would be beneficial to more robustly pinpoint the relationship and account for the variation in opportunity based on a more localized environment. With city- or census tract- level data, future work can better examine the characteristics of place and its resources that are connected to improving access to supports for families experiencing economic hardship.

Conclusion

Participation in safety net programs has been proven beneficial to mitigate the harmful effects of childhood poverty. Yet, there is incomplete take-up. Ensuring children have access to these essential supports is important to reduce the barriers to opportunity and ensure children have an opportunity to thrive. This paper highlights the importance of placed-based resources to reach families experiencing economic hardship. Agencies can improve outreach through partnerships with institutions that are critical resources for the target population. In communities with greater cohesion and where families rely on institutional resources, these institutions may be better suited to reach more vulnerable populations and broker connections to broader federal programs that promote child welfare. Even in larger urban areas, there are critical resources that families rely on. With individual or census level data, future research can identify the key players in these larger areas to improve access for families experiencing economic hardship.

References

- Bertrand, M., Luttmer, E. F. P., & Mullainathan, S. (2000). Network Effects and Welfare Cultures*. *The Quarterly Journal of Economics*, 115(3), 1019–1055. <https://doi.org/10.1162/003355300554971>
- Bradley, R. H., Corwyn, R. F., Burchinal, M., McAadoo, H. P., & García Coll, C. (2001). The Home Environments of Children in the United States Part II: Relations with Behavioral Development through Age Thirteen. *Child Development*, 72(6), 1868–1886. <https://doi.org/10.1111/1467-8624.t01-1-00383>
- Brooks-Gunn, J., & Duncan, G. J. (1997). The Effects of Poverty on Children. *The Future of Children*, 7(2), 55–71. <https://doi.org/10.2307/1602387>
- Chaudry, A., & Wimer, C. (2016). Poverty is Not Just an Indicator: The Relationship Between Income, Poverty, and Child Well-Being. *Academic Pediatrics, Child Poverty in the United States*, 16(3, Supplement), S23–S29. <https://doi.org/10.1016/j.acap.2015.12.010>
- Chetty, R., Hendren, N., & Katz, L. F. (2016). The Effects of Exposure to Better Neighborhoods on Children: New Evidence from the Moving to Opportunity Experiment. *American Economic Review*, 106(4), 855–902. <https://doi.org/10.1257/aer.20150572>
- Domina, T., Clark, L., Radskey, V., & Bhaskar, R. (2024). There Is Such a Thing as a Free Lunch: School Meals, Stigma, and Student Discipline. *American Educational Research Journal*, 61(2), 287–327. <https://doi.org/10.3102/00028312231222266>
- Dreyer, B. P. (2019). Safety Net Policies, Child Poverty, and Development Across the Lifespan. *Annual Review of Developmental Psychology*, 1(Volume 1, 2019), 337–357. <https://doi.org/10.1146/annurev-devpsych-121318-084855>
- Duncan, G. J., Morris, P. A., & Rodrigues, C. (2011). Does money really matter? Estimating impacts of family income on young children's achievement with data from random-assignment experiments. *Developmental Psychology*, 47(5), 1263–1279. <https://doi.org/10.1037/a0023875>
- Duncan, G. J., & Murnane, R. J. (2016). Rising Inequality in Family Incomes and Children's Educational Outcomes. *RSF: The Russell Sage Foundation Journal of the Social Sciences*, 2(2), 142–158. <https://doi.org/10.7758/rsf.2016.2.2.06>
- Evans, G. W. (2004). The Environment of Childhood Poverty. *American Psychologist*, 59(2), 77–92. <https://doi.org/10.1037/0003-066X.59.2.77>
- Frongillo, E. A., Jyoti, D. F., & Jones, S. J. (2006). Food Stamp Program Participation Is Associated with Better Academic Learning among School Children^{1,2}. *The Journal of Nutrition*, 136(4), 1077–1080. <https://doi.org/10.1093/jn/136.4.1077>
- Gassman-Pines, A., & Hill, Z. (2013). How Social Safety Net Programs Affect Family Economic Well-Being, Family Functioning, and Children's Development. *Child Development Perspectives*, 7(3), 172–181. <https://doi.org/10.1111/cdep.12037>
- Heckman, J. J., & Masterov, D. V. (2007). The Productivity Argument for Investing in Young Children*. *Applied Economic Perspectives and Policy*, 29(3), 446–493. <https://doi.org/10.1111/j.1467-9353.2007.00359.x>
- Herd, P., & Moynihan, D. P. (2019). *Administrative Burden: Policymaking by Other Means*. Russell Sage Foundation.
- Hernán, M. A., Brumback, B. A., & Robins, J. M. (2002). Estimating the causal effect of zidovudine on CD4 count with a marginal structural model for repeated measures. *Statistics in Medicine*, 21(12), 1689–1709. <https://doi.org/10.1002/sim.1144>

- Holland, P. W. (1986). Statistics and Causal Inference. *Journal of the American Statistical Association*, 81(396), 945–960. <https://doi.org/10.1080/01621459.1986.10478354>
- Homonoff, T., & Somerville, J. (2021). Program Recertification Costs: Evidence from SNAP. *American Economic Journal: Economic Policy*, 13(4), 271–298. <https://doi.org/10.1257/pol.20190272>
- Hong, G., & Raudenbush, S. W. (2008). Causal Inference for Time-Varying Instructional Treatments. *Journal of Educational and Behavioral Statistics*, 33(3), 333–362. <https://doi.org/10.3102/1076998607307355>
- Hoynes, H., Schanzenbach, D. W., & Almond, D. (2016). Long-Run Impacts of Childhood Access to the Safety Net. *American Economic Review*, 106(4), 903–934. <https://doi.org/10.1257/aer.20130375>
- Jackson, C. K., & Mackevicius, C. L. (2024). What Impacts Can We Expect from School Spending Policy? Evidence from Evaluations in the United States. *American Economic Journal: Applied Economics*, 16(1), 412–446. <https://doi.org/10.1257/app.20220279>
- Jencks, C., & Mayer, S. E. (1990). The Social Consequences of Growing up in a poor neighborhood. In *Inner-City Poverty in the United States* (Vol. 111, p. 186). National Academies Press.
- Robins, J. M., Hernán, M. Á., & Brumback, B. (2000). Marginal Structural Models and Causal Inference in Epidemiology. *Epidemiology*, 11(5), 550.
- Sampson, R. J., Sharkey, P., & Raudenbush, S. W. (2008). Durable effects of concentrated disadvantage on verbal ability among African-American children. *Proceedings of the National Academy of Sciences*, 105(3), 845–852. <https://doi.org/10.1073/pnas.0710189104>
- Schwartz, A. E., & Rothbart, M. W. (2020). Let Them Eat Lunch: The Impact of Universal Free Meals on Student Performance. *Journal of Policy Analysis and Management*, 39(2), 376–410. <https://doi.org/10.1002/pam.22175>
- Small, M. L. (2006). Neighborhood Institutions as Resource Brokers: Childcare Centers, Interorganizational Ties, and Resource Access among the Poor. *Social Problems*, 53(2), 274–292. <https://doi.org/10.1525/sp.2006.53.2.274>
- Small, M. L., & Gose, L. E. (2020). How Do Low-Income People Form Survival Networks? Routine Organizations as Brokers. *The ANNALS of the American Academy of Political and Social Science*, 689(1), 89–109. <https://doi.org/10.1177/0002716220915431>
- Stovel, K., & Shaw, L. (2012). Brokerage. *Annual Review of Sociology*, 38(Volume 38, 2012), 139–158. <https://doi.org/10.1146/annurev-soc-081309-150054>
- Warren, M. (2005). Communities and Schools: A New View of Urban Education Reform. *Harvard Educational Review*, 75(2), 133–173. <https://doi.org/10.17763/haer.75.2.m718151032167438>
- Wilson, W. J. (1997). *The Truly Disadvantaged: The Inner City, the Underclass, and Public Policy*. University of Chicago Press.
- Wodtke, G. T., Harding, D. J., & Elwert, F. (2011). Neighborhood Effects in Temporal Perspective: The Impact of Long-Term Exposure to Concentrated Disadvantage on High School Graduation. *American Sociological Review*, 76(5), 713–736. <https://doi.org/10.1177/0003122411420816>

Appendix

Figure 1A.
Community Resources per Child as a Continuous Measure, by Year



Note: Resource Level is defined as the sum of all resource organizations compared to the number of children under 6. Levels were created from the yearly distribution. *Source:* Business locations are from Data Axle and NCES Common Core

Table 1A.
Descriptive Statistics of Analysis Counties vs Others

	Analytic Sample	
	0	1
Churches	51.30 (32.58)	220.68 (148.29)
Preschools	11.15 (6.33)	58.14 (62.18)
Community Centers	1.30 (0.68)	7.69 (9.53)
Food Banks	1.00 (0.00)	1.76 (1.04)
Elementary Schools	5.67 (2.86)	26.64 (23.27)
Resources per Child	0.04 (0.01)	0.03 (0.01)
Unemployment Rate	5.03 (1.24)	4.49 (0.92)
Black	0.24 (0.19)	0.19 (0.11)
Hispanic	0.20 (0.03)	0.22 (0.02)
Poverty Percent, All Ages	18.11 (4.73)	14.77 (4.07)
Children below FPL (%)	32.30 (8.90)	25.01 (7.67)
Population	32,893 (18,044)	208,975 (222,780))

Note: 40 counties are included in the analytic sample from 2016 - 2019. The non-analytic sample includes 60 counties. *Source:* Participation data from NC DHHS. County Characteristics from ACS and USDA. Business Counts from Data Axle and NCES Common Core Data